A Finite-State Discount-to-Average Theorem for Rectangular Convex Risk-Averse MDPs

Partial progress on the Cheng–Jaimungal discount-to-one question

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Abstract

This note gives a rigorous finite-state theorem for the discount-to-average question raised in Remark 2.12 of Cheng and Jaimungal's distributional dynamic-risk-measure formulation of risk-averse Markov decision processes. The result is *partial progress*, not a solution of the full problem in that paper. The theorem applies to stationary finite-state models whose one-step risk transition maps admit rectangular compact convex-dual representations and whose dual transition envelopes satisfy a uniform minorization condition. Under these assumptions the discounted value vector v_γ satisfies

$$\lim_{\gamma\uparrow 1}(1-\gamma)v_\gamma(i) = \lambda, \qquad i \in \mathsf{S},$$

where λ is the unique additive eigenvalue in the average-risk Bellman equation

$$\lambda + h(i) = \min_{a\in A(i)} \sup_{q\in\mathcal{Q}_{ia}} \left\{ \sum_j q_j\big(c(i,a,j)+h(j)\big) - \alpha_{ia}(q) \right\}.$$

The same scalar λ is the optimal long-run average value of the corresponding nested rectangular dynamic risk criterion. In the coherent case the average criterion is a robust average cost. In the noncoherent convex case it is the average of the dual risk-adjusted one-period costs $c - \alpha$.

1 Mathematical status and scope

Cheng and Jaimungal [2] develop risk-averse Markov decision processes using distributional dynamic risk measures. Their setting allows, among other features, general state and action spaces, randomized actions, latent costs, weakly continuous transition kernels, state-dependent law-invariant convex risk measures, and finite or infinite horizon criteria. Remark 2.12 of their paper points out that in the risk-neutral discrete setting the properly scaled limit as the discount factor tends to one coincides with average optimality, while the corresponding equivalence in their risk-averse distributional setting remains unclear.

The present note does *not* solve that general question. It proves a narrower theorem under additional assumptions:

(i) the state space and the primitive action sets are finite;

(ii) the model is stationary;

(iii) the one-step risk transition maps are specified directly by rectangular dual envelopes over next-state distributions;

(iv) the envelopes satisfy a common Doeblin-type minorization condition.

Within this narrower class the discount-to-average statement is fully proved, including existence of the average Bellman eigenpair, convergence of scaled discounted values, and average-risk optimality of a stationary selector. Thus the result should be read as a sufficient-condition theorem and as partial progress on Remark 2.12, not as a resolution of the full Cheng–Jaimungal problem.

2 Finite stationary rectangular-dual model

Let

$$\mathsf{S} = \{1, \dots, m\}$$

be a finite state space. In state i, the controller chooses an action $a \in A(i)$, where each nonempty set $A(i)$ is finite. The one-period cost incurred when the current state is i, the action is a, and the next state is j is denoted by $c(i, a, j) \in \mathbb{R}$. Since all sets are finite, c is bounded.

For each state-action pair (i, a), let

$$\rho_{ia} : \mathbb{R}^m \to \mathbb{R}$$

be the one-step risk transition map applied to a vector of next-state continuation costs. We impose the following finite-dimensional rectangular dual representation.

Assumption 2.1 (Rectangular convex-dual representation)**.** *For each (i, a) there are a nonempty compact set $\mathcal{Q}_{ia} \subseteq \Delta(\mathsf{S})$ and a continuous penalty function*

$$\alpha_{ia} : \mathcal{Q}_{ia} \to [0, \infty)$$

such that

$$\rho_{ia}(z) = \sup_{q \in \mathcal{Q}_{ia}} \{q \cdot z - \alpha_{ia}(q)\}, \qquad z \in \mathbb{R}^m. \tag{1}$$

We also assume normalization, in the form

$$\min_{q \in \mathcal{Q}_{ia}} \alpha_{ia}(q) = 0, \qquad i \in \mathsf{S},\ a \in A(i).$$

The compactness and continuity assumptions guarantee attainment of all displayed suprema. Formula (1) is not automatic in the full Cheng–Jaimungal framework; it is an additional finite-dimensional rectangularity assumption. If randomized actions are to be included, they must be included as part of the action variable and one must reverify Assumption 2.1 for the induced one-step distribution of cost and next state. The theorem below does not assert that this reduction is automatic.

Define the undiscounted Bellman operator $T : \mathbb{R}^m \to \mathbb{R}^m$ by

$$(Tv)(i) = \min_{a \in A(i)} \rho_{ia}\big(c(i, a, \cdot) + v(\cdot)\big). \tag{2}$$

Equivalently,

$$(Tv)(i) = \min_{a \in A(i)} \sup_{q \in \mathcal{Q}_{ia}} \left\{ \sum_{j=1}^{m} q_j \big(c(i, a, j) + v(j)\big) - \alpha_{ia}(q) \right\}. \tag{3}$$

For $\gamma \in (0, 1)$ the discounted Bellman equation is

$$v_\gamma = T(\gamma v_\gamma), \tag{4}$$

that is,

$$v_\gamma(i) = \min_{a\in A(i)} \sup_{q\in\mathcal{Q}_{ia}} \left\{ \sum_{j=1}^{m} q_j\big(c(i,a,j) + \gamma v_\gamma(j)\big) - \alpha_{ia}(q) \right\}. \tag{5}$$

For $z \in \mathbb{R}^m$ write

$$\operatorname{osc}(z) = \max_i z(i) - \min_i z(i).$$

3 A communicating condition

The following assumption is a finite-dimensional nonlinear analogue of a strong communicating condition. It is sufficient, not necessary.

Assumption 3.1 (Uniform dual minorization)**.** *There exist a probability vector $\nu \in \Delta(\mathsf{S})$ and a number $\delta \in (0,1]$ such that*

$$q \ge \delta\nu \quad \textit{componentwise for every } i \in \mathsf{S},\ a \in A(i),\ q \in \mathcal{Q}_{ia}. \tag{6}$$

Equivalently, every admissible dual transition can be decomposed as

$$q = \delta\nu + (1-\delta)r$$

for some $r \in \Delta(\mathsf{S})$.

Remark 3.2. Assumption 3.1 excludes many interesting models. For example, if an envelope contains all point masses on a state space with at least two states, no full-support minorization can hold. The role of the assumption is to provide a clean checkable condition under which the Bellman operator has a unique additive eigenvalue. The Abelian part of the proof only needs existence of such an eigenpair.

4 Main theorem

The theorem combines three claims: the average Bellman eigenpair exists, scaled discounted values converge to its eigenvalue, and this eigenvalue is the long-run average value of the nested rectangular risk criterion.

Theorem 4.1 (Finite-state discount-to-average theorem)**.** *Assume Assumptions 2.1 and 3.1. Then:*

(a) There are $\lambda \in \mathbb{R}$ and $h \in \mathbb{R}^m$ satisfying the additive Bellman equation

$$Th = h + \lambda\mathbf{1}. \tag{7}$$

The scalar λ is unique, and h is unique up to addition of a constant.

(b) For every $\gamma \in (0,1)$ the discounted equation (4) has a unique solution v_γ, and

$$\lim_{\gamma\uparrow 1}(1-\gamma)v_\gamma(i) = \lambda, \qquad i \in \mathsf{S}. \tag{8}$$

More precisely, for every eigenvector h in (7),

$$\max_i |(1-\gamma)v_\gamma(i) - \lambda| \le 2(1-\gamma)\|h\|_\infty. \tag{9}$$

(c) *Let $g^*(i)$ be the optimal average value of the nested rectangular risk criterion defined in Section 7. Then*

$$g^*(i) = \lambda, \qquad i \in \mathsf{S}. \tag{10}$$

Moreover, every stationary deterministic selector a^* attaining the minimum in

$$\lambda + h(i) = \min_{a \in A(i)} \sup_{q \in \mathcal{Q}_{ia}} \left\{ \sum_j q_j \big(c(i,a,j) + h(j)\big) - \alpha_{ia}(q) \right\} \tag{11}$$

is average-risk optimal.

Consequently,

$$\lim_{\gamma \uparrow 1} (1-\gamma) v_\gamma(i) = g^*(i) = \lambda, \qquad i \in \mathsf{S}. \tag{12}$$

The proof is given in the next three sections.

5 Operator estimates and the additive eigenpair

Lemma 5.1 (Basic properties of T)**.** *The operator T is monotone and additively homogeneous:*

$$u \le v \implies Tu \le Tv, \qquad T(v + r\mathbf{1}) = Tv + r\mathbf{1} \quad (r \in \mathbb{R}).$$

It is nonexpansive in the sup norm:

$$\|Tu - Tv\|_\infty \le \|u - v\|_\infty. \tag{13}$$

Under Assumption 3.1, it is a contraction in the span seminorm:

$$\operatorname{osc}(Tu - Tv) \le (1-\delta) \operatorname{osc}(u - v), \qquad u, v \in \mathbb{R}^m. \tag{14}$$

Proof. Monotonicity follows from $q \in \Delta(\mathsf{S})$ in (1). Additive homogeneity follows from $q \cdot \mathbf{1} = 1$.

Let $d = u - v$. For a fixed state i, choose an action a_v attaining the minimum in $(Tv)(i)$. Then

$$\begin{aligned}(Tu)(i) - (Tv)(i) &\le \sup_{q \in \mathcal{Q}_{ia_v}} \{q \cdot (c_{ia_v} + u) - \alpha_{ia_v}(q)\} - \sup_{q \in \mathcal{Q}_{ia_v}} \{q \cdot (c_{ia_v} + v) - \alpha_{ia_v}(q)\} \\ &\le \sup_{q \in \mathcal{Q}_{ia_v}} q \cdot d \le \max_j d(j),\end{aligned}$$

where $c_{ia}(j) = c(i,a,j)$. Exchanging u and v gives

$$(Tu)(i) - (Tv)(i) \ge \min_j d(j).$$

Thus each component of $Tu - Tv$ lies between $\min_j d(j)$ and $\max_j d(j)$, which implies (13).

For the span estimate, the preceding argument gives, for all states i, k,

$$(Tu)(i) - (Tv)(i) - \big((Tu)(k) - (Tv)(k)\big) \le \sup_{q, q' \in \mathfrak{Q}} (q - q') \cdot d,$$

where $\mathfrak{Q} = \bigcup_{i,a} \mathcal{Q}_{ia}$. By Assumption 3.1, write

$$q = \delta\nu + (1-\delta) r, \qquad q' = \delta\nu + (1-\delta) r'$$

with $r, r' \in \Delta(\mathsf{S})$. Then

$$(q - q') \cdot d = (1-\delta)(r - r') \cdot d \le (1-\delta) \operatorname{osc}(d).$$

Taking the supremum over i, k proves (14). $\square$

Proposition 5.2 (Existence and uniqueness of the additive eigenpair)**.** *Under Assumptions 2.1 and 3.1, there exist $\lambda \in \mathbb{R}$ and $h \in \mathbb{R}^m$ such that $Th = h + \lambda\mathbf{1}$. The scalar λ is unique and h is unique modulo constants.*

Proof. Fix a reference state i_0. Let

$$B = \{v \in \mathbb{R}^m : v(i_0) = 0\},$$

equipped with the norm $\operatorname{osc}(\cdot)$. Define

$$F(v) = Tv - (Tv)(i_0)\mathbf{1}, \qquad v \in B.$$

Then F maps B into itself. By Lemma 5.1,

$$\operatorname{osc}(F(u) - F(v)) = \operatorname{osc}(Tu - Tv) \le (1-\delta)\operatorname{osc}(u - v).$$

Thus F is a contraction on the complete finite-dimensional normed space B. Banach's fixed point theorem gives a unique $h \in B$ satisfying $F(h) = h$. Put

$$\lambda = (Th)(i_0).$$

Then $Th = h + \lambda\mathbf{1}$.

Suppose $Tg = g + \mu\mathbf{1}$ is another additive eigenpair. Then

$$\operatorname{osc}(g - h) = \operatorname{osc}(Tg - Th) \le (1-\delta)\operatorname{osc}(g - h).$$

Since $\delta > 0$, this forces $\operatorname{osc}(g - h) = 0$, so $g - h$ is constant. Substitution into the two eigen-equations gives $\mu = \lambda$. □

6 Discounted values and Abelian convergence

Lemma 6.1 (Discounted fixed point)**.** *For every $\gamma \in (0, 1)$, the discounted Bellman equation $v = T(\gamma v)$ has a unique solution $v_\gamma \in \mathbb{R}^m$.*

Proof. The map $v \mapsto T(\gamma v)$ is a γ-contraction in the sup norm by Lemma 5.1. Banach's fixed point theorem gives existence and uniqueness. □

Proposition 6.2 (Abelian estimate)**.** *Suppose that T has an additive eigenpair (λ, h), not necessarily obtained from Assumption 3.1. Let v_γ be the unique solution of $v = T(\gamma v)$. Then*

$$\max_i |(1-\gamma)v_\gamma(i) - \lambda| \le 2(1-\gamma)\|h\|_\infty.$$

In particular $(1-\gamma)v_\gamma(i) \to \lambda$ for every state i as $\gamma \uparrow 1$.

Proof. Let $H = \|h\|_\infty$ and set

$$d_\gamma = v_\gamma - h, \qquad m_\gamma = \min_i d_\gamma(i), \qquad M_\gamma = \max_i d_\gamma(i).$$

Since

$$h + m_\gamma\mathbf{1} \le v_\gamma \le h + M_\gamma\mathbf{1},$$

monotonicity and additive homogeneity of T imply

$$v_\gamma = T(\gamma v_\gamma) \le T(\gamma h + \gamma M_\gamma \mathbf{1}) = T(\gamma h) + \gamma M_\gamma \mathbf{1},$$
$$v_\gamma = T(\gamma v_\gamma) \ge T(\gamma h + \gamma m_\gamma \mathbf{1}) = T(\gamma h) + \gamma m_\gamma \mathbf{1}.$$

By sup-norm nonexpansiveness,

$$\|T(\gamma h) - T(h)\|_\infty \le \|(\gamma - 1)h\|_\infty = (1-\gamma)H.$$

Since $T(h) = h + \lambda\mathbf{1}$, we obtain

$$v_\gamma \le h + \lambda\mathbf{1} + (1-\gamma)H\mathbf{1} + \gamma M_\gamma \mathbf{1},$$
$$v_\gamma \ge h + \lambda\mathbf{1} - (1-\gamma)H\mathbf{1} + \gamma m_\gamma \mathbf{1}.$$

Subtracting h and taking the maximum in the first inequality and the minimum in the second yields

$$(1-\gamma)M_\gamma \le \lambda + (1-\gamma)H, \qquad (1-\gamma)m_\gamma \ge \lambda - (1-\gamma)H.$$

For every state i,

$$(1-\gamma)v_\gamma(i) = (1-\gamma)h(i) + (1-\gamma)d_\gamma(i),$$

and $m_\gamma \le d_\gamma(i) \le M_\gamma$. Therefore

$$\lambda - 2(1-\gamma)H \le (1-\gamma)v_\gamma(i) \le \lambda + 2(1-\gamma)H,$$

which is the claimed estimate. □

7 Average-risk criterion and optimality

We now define the average criterion to which the discounted problem converges. This criterion is part of the present finite rectangular-dual model; it is not asserted to be the only possible average-risk formulation in the full Cheng–Jaimungal setting.

A deterministic history-dependent policy is a sequence $\pi = (\pi_t)_{t\ge 0}$ such that π_t maps each finite history

$$H_t = (X_0, A_0, X_1, A_1, \dots, X_t)$$

to an action $A_t = \pi_t(H_t) \in A(X_t)$. Let Π denote the class of all such policies.

For a finite horizon N and a policy π, define the nested risk value backward by $R^\pi_{N,N} = 0$ and

$$R^\pi_{t,N}(H_t) = \rho_{X_t,A_t}\Big(c(X_t, A_t, \cdot) + R^\pi_{t+1,N}(H_t, A_t, \cdot)\Big), \qquad t = N-1, \dots, 0. \tag{15}$$

Here $R^\pi_{t+1,N}(H_t, A_t, j)$ means the continuation value at the extended history in which $X_{t+1} = j$. Define

$$\mathcal{R}^\pi_N(i) = R^\pi_{0,N}(X_0 = i).$$

The average-risk value of π from state i is

$$\bar{J}(i,\pi) = \limsup_{N\to\infty} \frac{1}{N}\mathcal{R}^\pi_N(i), \tag{16}$$

and the optimal average-risk value is

$$g^*(i) = \inf_{\pi\in\Pi} \bar{J}(i,\pi). \tag{17}$$

The next lemma identifies the rectangular robust form of the nested risk value. An adversarial kernel is a history-dependent choice of a vector

$$q_t(H_t) \in \mathcal{Q}_{X_t,A_t}$$

after observing the current history and the controller's action. For fixed π and $q = (q_t)$, let $\mathbb{E}_i^{\pi,q}$ denote expectation for the controlled process starting at i and using transition probabilities $q_t(H_t)$.

Lemma 7.1 (Finite-horizon rectangular dual representation)**.** *For every deterministic history-dependent policy π and horizon N,*

$$\mathcal{R}_N^{\pi}(i) = \sup_q \mathbb{E}_i^{\pi,q}\left[\sum_{t=0}^{N-1}\Big(c(X_t, A_t, X_{t+1}) - \alpha_{X_t,A_t}(q_t(H_t))\Big)\right], \tag{18}$$

where the supremum is over all history-dependent adversarial kernels $q_t(H_t) \in \mathcal{Q}_{X_t,A_t}$.

Proof. The proof is backward induction on t. The assertion is trivial at time N. Suppose it holds from time $t+1$ onward. Fix a history H_t and write $i = X_t$, $a = A_t$. By Assumption 2.1,

$$R_{t,N}^{\pi}(H_t) = \sup_{q\in\mathcal{Q}_{ia}}\left\{\sum_j q_j\big(c(i,a,j) + R_{t+1,N}^{\pi}(H_t,a,j)\big) - \alpha_{ia}(q)\right\}.$$

For each chosen first-step q, the induction hypothesis represents $R_{t+1,N}^{\pi}(H_t,a,j)$ as a supremum over future adversarial kernels. Because the uncertainty is rectangular, the first-step maximizer and the future kernels can be chosen independently after each realized next state. This yields the supremum over all kernels from time t to $N-1$ of the conditional expected sum of $c-\alpha$. Taking $t = 0$ gives (18). $\square$

Proposition 7.2 (Average optimality equation)**.** *Suppose that T has an additive eigenpair (λ, h). Then $g^*(i) = \lambda$ for every $i \in \mathsf{S}$. Moreover, any stationary selector a^* attaining the minimum in (11) is average-risk optimal.*

Proof. Let a^* be a selector attaining the minimum in (11), and let π^* be the stationary policy $A_t = a^*(X_t)$. From the eigen-equation, for every state i and every $q \in \mathcal{Q}_{i,a^*(i)}$,

$$\sum_j q_j\big(c(i,a^*(i),j) + h(j)\big) - \alpha_{i,a^*(i)}(q) \le \lambda + h(i). \tag{19}$$

For any adversarial kernel $q = (q_t)$, conditioning on the history H_t gives

$$\mathbb{E}_i^{\pi^*,q}\big[c(X_t,A_t,X_{t+1}) - \alpha_{X_t,A_t}(q_t(H_t)) + h(X_{t+1}) \mid H_t\big] \le \lambda + h(X_t).$$

Summing from $t=0$ to $N-1$ yields

$$\mathbb{E}_i^{\pi^*,q}\left[\sum_{t=0}^{N-1}\big(c(X_t,A_t,X_{t+1}) - \alpha_{X_t,A_t}(q_t(H_t))\big)\right] \le N\lambda + h(i) - \mathbb{E}_i^{\pi^*,q}h(X_N).$$

Since h is bounded, the right side is at most $N\lambda + \operatorname{osc}(h)$. Taking the supremum over q and applying Lemma 7.1,

$$\mathcal{R}_N^{\pi^*}(i) \le N\lambda + \operatorname{osc}(h).$$

Therefore $\bar{J}(i,\pi^*) \le \lambda$.

Conversely, fix an arbitrary policy $\pi \in \Pi$. Since

$$\lambda + h(i) = \min_{a \in A(i)} \sup_{q \in \mathcal{Q}_{ia}} \left\{ \sum_j q_j \big(c(i,a,j) + h(j)\big) - \alpha_{ia}(q) \right\},$$

we have, for every state-action pair (i,a),

$$\sup_{q \in \mathcal{Q}_{ia}} \left\{ \sum_j q_j \big(c(i,a,j) + h(j)\big) - \alpha_{ia}(q) \right\} \geq \lambda + h(i).$$

The supremum is attained. Choose one maximizer $q^*(i,a)$ for each state-action pair. Let the adversary use $q_t(H_t) = q^*(X_t, A_t)$. Then

$$\mathbb{E}_i^{\pi,q^*}[c(X_t, A_t, X_{t+1}) - \alpha_{X_t,A_t}(q_t(H_t)) + h(X_{t+1}) \mid H_t] \geq \lambda + h(X_t).$$

Summing gives

$$\mathbb{E}_i^{\pi,q^*}\left[\sum_{t=0}^{N-1} \big(c(X_t, A_t, X_{t+1}) - \alpha_{X_t,A_t}(q_t(H_t))\big)\right] \geq N\lambda + h(i) - \mathbb{E}_i^{\pi,q^*} h(X_N) \geq N\lambda - \operatorname{osc}(h).$$

Using Lemma 7.1,

$$\mathcal{R}_N^\pi(i) \geq N\lambda - \operatorname{osc}(h).$$

Thus $\bar{J}(i,\pi) \geq \lambda$ for every policy π. Taking the infimum over π gives $g^*(i) \geq \lambda$. Together with $\bar{J}(i,\pi^*) \leq \lambda$, this proves $g^*(i) = \lambda$ and the optimality of π^*. □

Proof of Theorem 4.1. Part (a) is Proposition 5.2. Part (b) is Lemma 6.1 together with Proposition 6.2. Part (c) is Proposition 7.2. Equation (12) follows by combining parts (b) and (c). □

8 Interpretation

8.1 Risk-neutral case

If $\rho_{ia}(z) = \sum_j P_{ij}^a z(j)$, then $\mathcal{Q}_{ia} = \{P_i^a\}$ and $\alpha_{ia} \equiv 0$. The eigenproblem becomes

$$\lambda + h(i) = \min_{a \in A(i)} \sum_j P_{ij}^a \big(c(i,a,j) + h(j)\big),$$

which is the classical average-cost optimality equation. Assumption 3.1 reduces to a common Doeblin lower bound on the transition laws P_i^a.

8.2 Coherent risk maps

For coherent rectangular risk maps, the penalty can be taken to vanish on the envelope. The eigenproblem is then

$$\lambda + h(i) = \min_{a \in A(i)} \sup_{q \in \mathcal{Q}_{ia}} \sum_j q_j \big(c(i,a,j) + h(j)\big),$$

and the average criterion in Lemma 7.1 is an ordinary robust long-run average cost.

8.3 Convex noncoherent risk maps

For convex noncoherent risk maps, the penalty α_{ia} remains in the eigenproblem and in the finite-horizon robust representation. Hence the limiting average object is generally not the robust average of c alone. It is the long-run average of the dual risk-adjusted costs

$$c(X_t, A_t, X_{t+1}) - \alpha_{X_t,A_t}(q_t).$$

This is forced by the nested dual representation; it is not an optional correction.

9 Why no unrestricted scalar theorem is possible

Even in the risk-neutral case, a state-independent scalar limit can fail without a communicating assumption.

Example 9.1 (Two absorbing classes)**.** *Let* $\mathsf{S} = \{1, 2\}$ *and suppose there is only one action. The transition matrix is the identity, and the one-period costs are* $c(1, 1) = 0$ *and* $c(2, 2) = 1$*. For ordinary expectation risk,*

$$v_\gamma(1) = 0, \qquad v_\gamma(2) = \frac{1}{1-\gamma}.$$

Thus

$$(1-\gamma)v_\gamma(1) \to 0, \qquad (1-\gamma)v_\gamma(2) \to 1.$$

There is no single average eigenvalue independent of the initial state. Therefore a theorem of the form proved above must either impose a communicating condition, allow state-dependent average classes, or use a different formulation.

10 Limitations relative to Cheng–Jaimungal

The theorem proves a genuine discount-to-average result, but only for a restrictive special case. The following points are the main limitations.

First, the model here is finite and stationary, while Cheng and Jaimungal allow much more general state and action spaces and time-dependent ingredients. Second, the theorem assumes a rectangular dual representation over next-state distributions. The distributional dynamic risk maps in [2] are defined through the law of the one-step cost plus continuation value; passing from that formulation to (1) requires additional finite-dimensional duality and rectangularity arguments. Third, randomized actions are not handled automatically by the proof. Nonlinear risk of a randomized one-step cost distribution need not reduce to simply mixing the deterministic-action Bellman expressions. Fourth, the uniform minorization condition is strong; it gives a clean contraction proof but is far from necessary and may fail for natural envelopes.

Thus the conclusion should be stated as follows.

> The finite stationary rectangular-dual Cheng–Jaimungal-type model has a positive discount-to-average theorem under a uniform dual minorization condition. This is partial progress on Remark 2.12. It is not a solution of the full general problem posed by the authors.

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